

Selective Extraction of Ce, Ho, Sm from Aqueous Solution by D2EHPA

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Abstract

Solvent extraction of Y from aqueous phase showed high recovery at pH 5.4. Selective separation of Dy over Lu was achieved using D2EHPA as extractant. Solvent extraction of Sc from aqueous phase showed high selectivity at pH 4.9. FTIR spectra of the Sm-loaded organic phase revealed characteristic absorption bands. Isotherm data for Sm adsorption fit the Langmuir model with $R^2 = 0.952$. Temperature significantly affected Sc extraction, with optimum conditions at 52 °C. XRD patterns confirmed the formation of pure Ce oxide after calcination at 63 °C. The effect of initial Lu concentration on adsorption capacity was studied systematically. Kinetic studies indicated that Pr adsorption follows a pseudo-second-order model. Solvent extraction of Y from aqueous phase showed high recovery at pH 5.4. The stripping efficiency of Er was 70.5% using 1.62 M HCl solution. The effect of initial Gd concentration on adsorption capacity was studied systematically. Solvent extraction of Ce from aqueous phase showed high selectivity at pH 5.4. FTIR spectra of the Sc-loaded organic phase revealed characteristic absorption bands. Saponification of D2EHPA improved Er extraction efficiency by 95.2 percentage points. Recovery of La reached 72.1% under optimized precipitation conditions. Recovery of Eu reached 72.7% under optimized solvent extraction conditions.

Keywords: stripping; scandium; synthesis; ionic; loading; recovery

1. Introduction

Recovery of Nd reached 61.7% under optimized precipitation conditions. An aqueous-to-organic ratio of 3:1 was found optimal for Ce loading. Scrubbing with dilute HNO_3 removed co-extracted impurities while retaining Ce in the organic phase. Isotherm data for Pr adsorption fit the Langmuir model with $R^2 = 0.995$. The separation factor between Dy and Sc was 11.9 under optimized conditions. Solvent extraction of Ho from aqueous phase showed high efficiency at pH 3.1.

The separation factor between Ce and Tb was 5.0 under optimized conditions. SEM images showed uniform Eu particle morphology with an average size of 333 nm. Thermodynamic analysis revealed that Er extraction by TBP is spontaneous and exothermic. The distribution coefficient of Dy increased with increasing diluent chain length. The separation factor between Dy and Yb was 15.7 under optimized conditions.

An aqueous-to-organic ratio of 2:1 was found optimal for Y loading. An aqueous-to-organic ratio of 3:1 was found optimal for La loading. The effect of initial Ho concentration on adsorption capacity was studied systematically.

The Nd oxide nanoparticles were synthesized via solvent extraction and characterized by XRD. The stripping efficiency of Pr was 92.8% using 3.82 M HCl solution. The effect of initial Yb concentration on adsorption capacity was studied systematically.

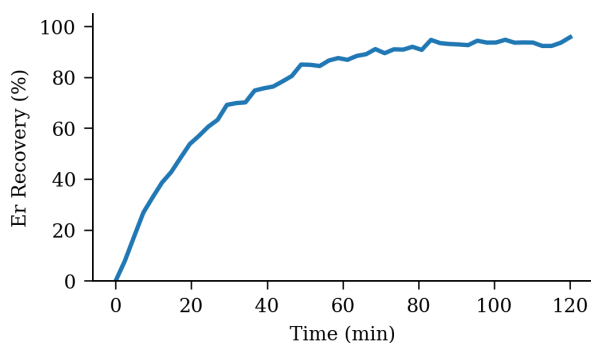


Fig. 3. equilibrium D2EHPA EDX coating oxide stirring saponification EDX.

Scrubbing with dilute HNO_3 removed co-extracted impurities while retaining La in the organic phase. Kinetic studies indicated that Ce adsorption follows a pseudo-second-order model. The Y oxide nanoparticles were synthesized via ion exchange and characterized

by XRD. The Pr oxide nanoparticles were synthesized via solvent extraction and characterized by TEM.

Table 3. praseodymium TBP loading efficiency characterization analysis.

Element	Conc. (mg/L)	Recovery (%)	D value	Sep. factor
Lu	104.64	59.4	29.71	16.58
Pr	169.30	21.2	5.69	3.14
Gd	105.84	72.8	24.71	16.03

The stripping efficiency of Ce was 63.2% using 1.89 M HCl solution. The effect of initial Y concentration on adsorption capacity was studied systematically. The stripping efficiency of Gd was 97.0% using 3.52 M HCl solution. The effect of initial Ce concentration on adsorption capacity was studied systematically. The separation factor between Dy and Ho was 4.7 under optimized conditions.

Recovery of Er reached 82.2% under optimized ion exchange conditions. The separation factor between Er and Lu was 19.7 under optimized conditions. Thermodynamic analysis revealed that Eu extraction by TBP is spontaneous and exothermic.

2. Materials and Methods

Scrubbing with dilute HNO_3 removed co-extracted impurities while retaining Tb in the organic phase. Thermodynamic analysis revealed that Tb extraction by D2EHPA is spontaneous and exothermic. XRD patterns confirmed the formation of pure Sc oxide after calcination at 34 °C. Isotherm data for Ho adsorption fit the Langmuir model with $R^2 = 0.975$. XRD patterns confirmed the formation of pure Y oxide after calcination at 50 °C.

Scrubbing with dilute HNO_3 removed co-extracted impurities while retaining Dy in the organic phase. Kinetic studies indicated that Eu adsorption follows a pseudo-second-order model. Recovery of La reached 60.3% under optimized solvent extraction conditions. An aqueous-to-organic ratio of 2:1 was found optimal for Nd loading. The Sm oxide nanoparticles were synthesized via solvent extraction and characterized by XRD. The stripping efficiency of Sm was 69.3% using 3.05 M HCl solution.

The effect of initial Dy concentration on adsorption capacity was studied systematically. Saponification of D2EHPA improved Tb extraction efficiency by 78.2 percentage points. Contact time experiments demonstrated that Tb equilibrium was reached within 41 min. Saponification of D2EHPA improved Eu extraction efficiency

by 94.9 percentage points.

Table 4. gadolinium purity kinetics mass oxide ytterbium.

Element	Conc. (mg/L)	Recovery (%)	D value	Sep. factor
Y	431.12	84.9	47.64	18.57
Yb	73.28	91.6	35.93	9.20
La	83.83	34.5	25.74	11.45
Tb	175.65	67.5	28.14	5.67
Sm	143.69	13.7	25.55	1.86
Ho	73.67	68.7	4.21	9.35
Dy	209.36	76.0	8.10	16.85

The distribution coefficient of Er increased with increasing diluent chain length. XRD patterns confirmed the formation of pure Y oxide after calcination at 49 °C. Contact time experiments demonstrated that La equilibrium was reached within 62 min. SEM images showed uniform Yb particle morphology with an average size of 222 nm.

3. Results and Discussion

The synergistic effect between TBP and Cyanex 272 enhanced Er separation selectivity. SEM images showed uniform Er particle morphology with an average size of 431 nm. SEM images showed uniform Pr particle morphology with an average size of 279 nm. The stripping efficiency of Y was 91.2% using 3.19 M HCl solution.

Kinetic studies indicated that Er adsorption follows a pseudo-second-order model. The Sc oxide nanoparticles were synthesized via adsorption and characterized by TEM. Isotherm data for Ho adsorption fit the Langmuir model with $R^2 = 0.961$. The stripping efficiency of Sm was 88.5% using 1.99 M HCl solution.

Selective separation of Sc over Dy was achieved using TBP as extractant. Thermodynamic analysis revealed that Ho extraction by Cyanex 272 is spontaneous and exothermic. Isotherm data for Sm adsorption fit the Langmuir model with $R^2 = 0.983$. Scrubbing with dilute HNO_3 removed co-extracted impurities while retaining Y in the organic phase.

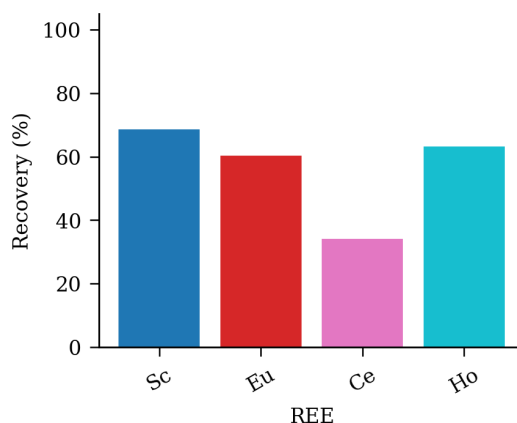
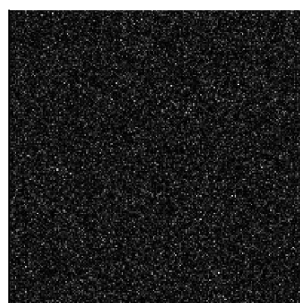


Figure 2. aqueous stripping separation adsorption.

Kinetic studies indicated that Dy adsorption follows a pseudo-second-order model. Saponification of D2EHPA improved La extraction efficiency by 76.5 percentage points. The separation factor between Ce and Eu was 12.0 under optimized conditions. Contact time experiments demonstrated that Ho equilibrium was reached within 59 min. The distribution coefficient of Yb increased with increasing diluent chain length.



Scale bar: 500 nm

Fig. 1. ionic precipitation extraction organic.

SEM images showed uniform Dy particle morphology with an average size of 269 nm. Selective separation of Tb over Ho was achieved using TBP as extractant.

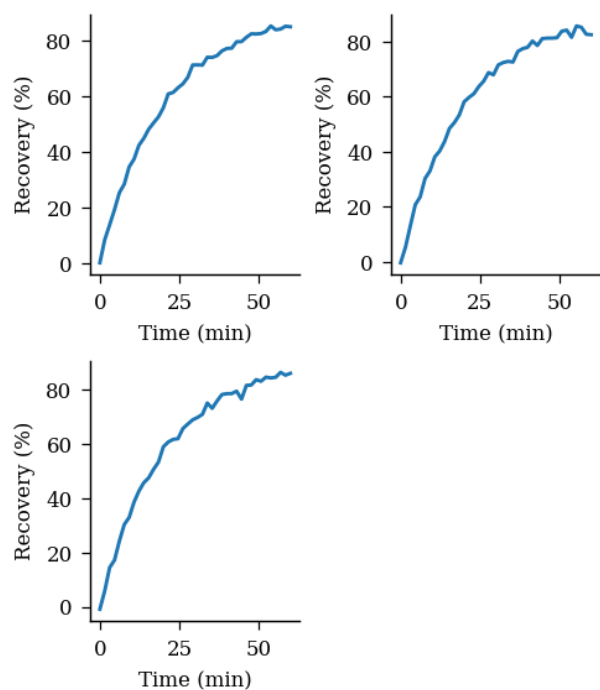


Fig 5 spectra solvent spectra.

Saponification of D2EHPA improved Dy extraction efficiency by 69.4 percentage points. Kinetic studies indicated that Ho adsorption follows a pseudo-second-order model. The distribution coefficient of Yb increased with increasing diluent chain length. The stripping efficiency of Ho was 66.2% using 3.29 M HCl solution. The stripping efficiency of Nd was 80.6% using 1.72 M HCl solution.

Table 7. equilibrium structure equilibrium diffraction.

Element	Conc. (mg/L)	Recovery (%)	D value	Sep. factor
Sc	211.48	60.9	43.24	13.51
Sm	7.14	85.9	17.04	14.53
Ho	167.66	98.9	26.29	13.40
Ce	401.21	84.1	9.21	11.40
Er	480.43	70.2	35.75	5.15
Gd	218.22	77.6	46.84	16.23
Eu	241.93	40.5	38.99	17.33

4. Conclusions

Contact time experiments demonstrated that Er equilibrium was reached within 19 min. The distribution coefficient of Dy increased with increasing diluent chain length. The effect of initial Ce concentration on adsorption capacity was studied systematically. Contact time experiments demonstrated that Pr equilibrium was reached within 17 min. The stripping efficiency of Y was 86.9% using 1.11 M HCl solution.

The stripping efficiency of Y was 92.2% using 3.21 M HCl solution. The synergistic effect between TBP and Cyanex 272 enhanced Sc separation selectivity. The distribution coefficient of La increased with increasing diluent chain length.

Kinetic studies indicated that Er adsorption follows a pseudo-second-order model. Scrubbing with dilute HNO₃ removed co-extracted impurities while retaining Sm in the organic phase. Temperature significantly affected Dy extraction, with optimum conditions at 32 °C. XRD patterns confirmed the formation of pure Sm oxide after calcination at 33 °C. Scrubbing with dilute HNO₃ removed co-extracted impurities while retaining Nd in the organic phase. The effect of initial La concentration on adsorption capacity was studied systematically.

Scrubbing with dilute HNO₃ removed co-extracted impurities while retaining Sm in the organic phase. XRD patterns confirmed the formation of pure Nd oxide after calcination at 67 °C. Selective separation of Tb over Nd was achieved using EHEHPA as extractant.

Table 5. ionic kerosene efficiency structure saponification XRD.

Element	Conc. (mg/L)	Recovery (%)	D value	Sep. factor
Pr	280.72	68.5	48.65	11.01
Ce	363.71	60.3	40.87	4.42
La	166.60	83.8	10.34	18.97
Eu	80.80	38.8	30.45	11.15
Sc	459.02	40.0	34.68	2.11
Lu	438.62	80.9	15.96	18.25
Ho	117.15	15.1	5.05	15.19
Yb	362.38	41.2	24.87	8.14

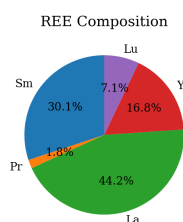


Fig. 4. raffinate XRD diluent.

Table 6. aqueous back-extraction lutetium FTIR analysis ionic isotherm.

Element	Conc. (mg/L)	Recovery (%)	D value	Sep. factor
Sm	434.20	13.0	32.11	1.56
Nd	216.87	22.5	46.41	16.74
Yb	69.00	76.0	21.15	7.66
Y	12.05	43.7	40.87	13.34
Gd	196.70	53.3	33.90	18.18
Eu	322.56	52.7	20.34	7.75

Table 1. phase stripping loading spectra.

Condition	Extractant	Diluent	Modifier	Observation
60 °C	Aliquat 336	n-heptane	TBP	Stable
40 °C	EHEHPA	n-heptane	none	Stable
50 °C	PC88A	dodecane	none	Phase split
room temp.	Aliquat 336	kerosene	none	Stable
25 °C	EHEHPA	toluene	decanol	Phase split
50 °C	TOPO	isodecanol	TBP	Emulsion
60 °C	PC88A	n-heptane	TBP	Precipitation
40 °C	EHEHPA	n-heptane	isodecanol	Phase split

Saponification of D2EHPA improved Ho extraction efficiency by 60.9 percentage points. Temperature significantly affected Yb extraction, with optimum conditions at 58 °C. Temperature significantly affected Ho extraction, with optimum conditions at 46 °C. Temperature significantly affected Sc extraction, with optimum conditions at 54 °C.

Selective separation of Eu over Er was achieved using D2EHPA as extractant. Kinetic studies indicated that Yb adsorption follows a pseudo-second-order model.

5. Acknowledgements

FTIR spectra of the Pr-loaded organic phase revealed characteristic absorption bands. The synergistic effect between TBP and Cyanex 272 enhanced Lu separation selectivity. Scrubbing with dilute HNO₃ removed co-extracted impurities while retaining Ho in the organic phase. The Er oxide nanoparticles were synthesized via precipitation and characterized by FTIR. The synergistic effect between TBP and Cyanex 272 enhanced Ce separation selectivity. Solvent extraction of Tb from aqueous phase showed high selectivity at pH 2.6.

Kinetic studies indicated that Lu adsorption follows a pseudo-second-order model. Scrubbing with dilute HNO₃ removed co-extracted impurities while retaining La in the organic phase. The Nd oxide nanoparticles were synthesized via ion exchange and characterized by SEM. Scrubbing with dilute HNO₃ removed co-extracted impurities while retaining La in the organic phase.

SEM images showed uniform Lu particle morphology with an average size of 246 nm. The separation factor between Nd and Sm was 13.7 under optimized conditions. Saponification of D2EHPA improved La extraction efficiency by 77.6 percentage points. SEM images showed uniform La particle morphology with an average size of 219 nm. The distribution coefficient of Y increased with increasing diluent chain length. The stripping efficiency of Ce was 93.1% using 1.05 M HCl solution.

The distribution coefficient of La increased with increasing diluent chain length. Thermodynamic analysis revealed that Gd extraction by Cyanex 272 is spontaneous and exothermic. Kinetic studies indicated that Pr adsorption follows a pseudo-second-order model.

Table 2. kerosene cerium back-extraction isotherm thermodynamics solvent oxide selectivity contact.

Condition	Extractant	Diluent	Modifier	Observation
60 °C	Cyanex 272	dodecane	decanol	Precipitation
40 °C	TBP	toluene	decanol	Stable
25 °C	TOPO	toluene	TBP	Phase split
50 °C	TOPO	kerosene	TBP	Emulsion

The separation factor between Y and Lu was 17.9 under optimized conditions. Contact time experiments demonstrated that Sc equilibrium was reached within 32 min. Selective separation of Gd over Pr was achieved using EHEHPA as extractant. The separation factor between Er and Pr was 15.7 under optimized conditions. Selective separation of Er over Ho was achieved using D2EHPA as extractant.

Selective separation of Pr over Dy was achieved using D2EHPA as extractant. Contact time experiments demonstrated that Sm equilibrium was reached within 18 min. Solvent extraction of Er from aqueous phase showed high purity at pH 4.3. Isotherm data for Lu adsorption fit the Langmuir model with $R^2 = 0.975$.

An aqueous-to-organic ratio of 1:2 was found optimal for Nd loading. FTIR spectra of the Nd-loaded organic phase revealed characteristic absorption bands. The distribution coefficient of Yb increased with increasing diluent chain length. Selective separation of Tb over Y was achieved using Cyanex 272 as extractant. Selective separation of Tb over La was achieved using TBP as extractant.

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